of this problem is a milestone in CT technology called the **Fourier slice theorem**.

Figure 25-18 shows how the problem looks in both the spatial and the frequency domains. In the spatial domain, each view is found by integrating the image along rays at a particular angle. In the frequency domain, the image spectrum is represented in this illustration by a two-dimensional grid. The spectrum of each view (a one-dimensional signal) is represented by a dark line superimposed on the grid. As shown by the positioning of the lines on the grid, the Fourier slice theorem states that the spectrum of a view is identical to the values along a line (slice) through the image spectrum. For instance, the spectrum of view 1 is the same as the center column of the image spectrum, and the spectrum of view 3 is the same as the center row of the image spectrum. Notice that the spectrum of each view is positioned on the grid at the same angle that the view was originally acquired. All these frequency spectra include the negative frequencies and are displayed with zero frequency at the center.

Fourier reconstruction of a CT image requires three steps. First, the one-dimensional FFT is taken of each view. Second, these view spectra are used to calculate the two-dimensional frequency spectrum of the image, as outlined by the Fourier slice theorem. Since the view spectra are arranged *radially*, and the correct image spectrum is arranged *rectangularly*, an interpolation routine is needed to make the conversion. Third, the inverse FFT is taken of the image spectrum to obtain the reconstructed image.

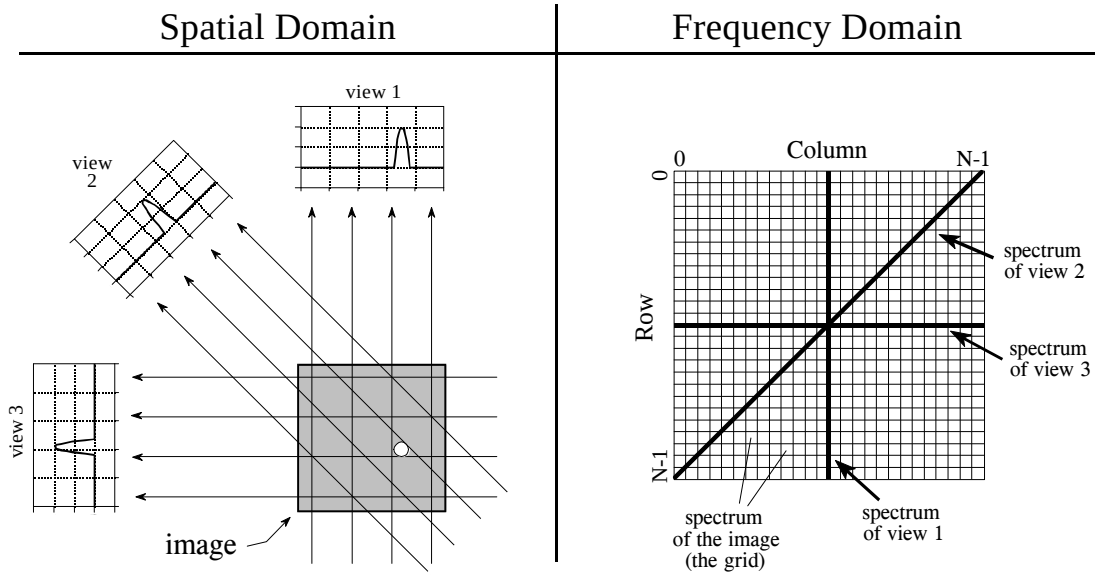

FIGURE 25-18
The Fourier Slice Theorem. The Fourier Slice Theorem describes the relationship between an image and its views in the frequency domain. In the spatial domain, each view is found by integrating the image along rays at a particular angle. In the frequency domain, the spectrum of each view is a one-dimensional "slice" of the two-dimensional image spectrum.